Role of hesperetin (a natural flavonoid) and its analogue on apoptosis in HT-29 human colon adenocarcinoma cell line--a comparative study.

Colon cancer is one of the serious health problems in most developed countries and its incidence rate is increasing in India. Hesperetin (HN) (3',5,7-trihydroxy-4'-methoxyflavonone) and hesperetin analogue (HA) were tested for their apoptosis inducing ability. Methyl thiazolyl tetrazolium assay revealed a dose as well as duration-dependent reduction of HT-29 (colon adenocarcinoma) cellular growth in response to HN and HA treatment. At 24 h 70 μM of HN and 32 μM of HA showed 50% reduction of HT-29 cellular growth. Acridine orange/ethidium bromide staining showed apoptotic features of cell death induced by HN and HA. Rhodamine 123 staining showed significant reduction in mitochondrial membrane potential induced by HN and HA. HN and HA induced DNA damage was confirmed by comet tail formation. Lipid peroxidation markers (TBARS) and protein oxidation marker (PCC) were significantly elevated in HN and HA treated groups. Enzymic antioxidants such as superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx) were slightly decreased in their activities compared to control (untreated HT-29 cells). Results of Western blot analysis of apoptosis associated genes revealed an increase in cytochrome C, Bax, cleaved caspase-3 expression and a decrease in Bcl-2 expression. These findings indicate that HN and HA induce apoptosis on HT-29 via Bax dependent mitochondrial pathway involving oxidant/antioxidant imbalance.